

PRESSURE VESSEL EXTERNAL INSPECTION REPORT

Crude Distillation Unit - Overhead Accumulator

Report No.: INS/2024/0117

Equipment Tag: V-1201

Unit: CDU-I

Date of Report: 18 Apr 2024

Service: Crude overhead accumulator

Inspected By: S. Rajagopal, Inspection Engineer

1. Scope of Inspection

This report covers the external in-service inspection of pressure vessel V-1201 carried out on 12 April 2024 during the unit run. The inspection comprised close visual examination of the shell, heads, nozzles, supports and attached piping, together with ultrasonic thickness measurement at the established condition monitoring locations.

The inspection was performed in accordance with API 510, Pressure Vessel Inspection Code, and the departmental procedure SOP-INSP-002. No internal inspection was carried out, the vessel being in service throughout.

2. Design and Construction Data

Parameter	Value	Unit	Source
Design pressure	10.5	barg	Data sheet DS-V-1201 Rev.3
Design temperature	165	deg C	Data sheet DS-V-1201 Rev.3
Operating pressure	8.2	barg	Process data
Shell material	SA-516 Gr.70	-	Data sheet DS-V-1201 Rev.3
Nominal shell thickness	14.0	mm	Fabrication drawing
Corrosion allowance	3.0	mm	Data sheet DS-V-1201 Rev.3
Joint efficiency	1.00	-	Radiography 100 percent
Year of commissioning	2009	-	Equipment register

3. Thickness Measurement Results

Ultrasonic thickness readings were taken at the twelve established condition monitoring locations using a calibrated digital thickness gauge. Readings are the minimum of five measurements taken within each location.

CML	Location	Nominal (mm)	Measured (mm)	Loss (mm)	Status
CML-01	Shell course 1, north	14.0	13.2	0.8	Acceptable
CML-02	Shell course 1, south	14.0	13.4	0.6	Acceptable
CML-03	Shell course 2, north	14.0	12.9	1.1	Acceptable
CML-04	Shell course 2, south	14.0	13.1	0.9	Acceptable
CML-05	Top head, crown	16.0	15.2	0.8	Acceptable
CML-06	Bottom head, crown	16.0	14.7	1.3	Acceptable
CML-07	Shell course 3, weld seam	14.0	11.4	2.6	Monitor
CML-08	Inlet nozzle N1 neck	12.0	11.1	0.9	Acceptable
CML-09	Outlet nozzle N2 neck	12.0	11.3	0.7	Acceptable
CML-10	Drain nozzle N4 neck	10.0	9.2	0.8	Acceptable
CML-11	Manway neck	14.0	13.5	0.5	Acceptable
CML-12	Skirt to shell junction	14.0	13.0	1.0	Acceptable

4. Visual Examination Findings

S.No	Observation	Severity	Recommended Action
1	Insulation cladding damaged at north nozzle N1, suspecte	Major	Strip insulation and inspect at next opportuni
2	Minor surface rust on skirt fireproofing	Minor	Touch up during next painting round
3	Support saddle grout intact, no cracking observed	Satisfactory	No action
4	Nameplate legible and securely attached	Satisfactory	No action
5	Localised thinning at CML-07 weld seam	Major	Increase inspection frequency to annual

5. Conclusion and Next Inspection

The vessel is fit for continued service at the current operating conditions. The remaining life governed by CML-07 is calculated as 9.2 years at the observed corrosion rate, giving a next internal inspection due date of March 2028. External inspection is to be repeated annually and the CML-07 location is to be added to the annual UT round.

Inspected By

Reviewed By

Approved By